

A Post-Quantum Relation-Bound Blind Certification Core with Signature-Gated Threshold Decryption

Research construction draft, version 2.5

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Abstract

This document specifies and proves an abstract post-quantum cryptographic core for privacy-preserving, accountable credentials. During an offline issuance phase, a holder obtains a blind signature only after proving that the hidden ticket, the blind-signing request, the holder secret, and a trace ciphertext are mutually consistent. During opening, threshold decryption shares are released only for a validly signed ticket and an independently authorized case. This *signature-gated threshold decryption* rule replaces a public validity proof carried by every online ticket.

The main result is conditional and modular: assuming simulation-extractable zero knowledge, one-more unforgeability and blindness of the selected blind-signature component, and a compact threshold encryption scheme with post-quantum privacy and unique openings, the construction provides issuer unlinkability, certified ciphertext validity, decoder safety, trace correctness, non-frameability, and a gated chosen-ciphertext privacy notion against fewer than the threshold number of opening authorities. The result deliberately does not claim ordinary IND-CCA security for the bare trace encryption. Version 1.2 also instantiates the trace relation as a Niederreiter syndrome plus a SHAKE256/KMAC256 one-time hash DEM, proves information-theoretic uniqueness of the decoded error, and removes a redundant ticket commitment. Version 1.3 freezes a paper-compatible Blind-UOV-Is/Shorter costing profile, fixes a 368-byte payload, gives a reproducible five-block circuit count, and separates the large issuance NIZK from the CAP proof carried by the final Blind-UOV signature. Version 1.4 adds an executable witness builder, a streaming characteristic-two circuit generator, independent SHAKE256/KMAC256 checks, and full-circuit rejection tests. Native Blind-UOV circuit integration, backend simulation extractability, and the threshold implementation remain explicit obligations. Version 1.5 lowers the complete incremental relation into a checksummed binary $\mathbb{F}_2$ -R1CS archive, independently parses every row, and separates the nonlinear-only cost model from the portable materialized-row count. Version 1.6 corrected the signer-view ABI but incorrectly treated two independently openable 256-bit request lanes as a 512-bit random-function output. Version 1.7 removes that unsound amplification. It selects one Blind-UOV-III/Shorter instance with a 576-bit request target and derives cross-message request binding in the QROM from CAP unique-witness soundness and the collision bound for a shifted 576-bit random function. The cached issuer key grows to 189.2 KB, while the online ticket remains 12,012 bytes. Native Blind-UOV circuit integration and proof-backend qualification remain explicit obligations. Version 1.8 writes $h_{\text{BUOV}} = H_{\text{BUOV}}(m, c_r)$ and internalizes the linear request equation $y = r + h_{\text{BUOV}}$ as 576 binary equalities. It narrows the sole external boundary to $h_{\text{BUOV}} = H_{\text{BUOV}}(m, \text{CAP.Commit}(r; \rho))$ and adds a fail-closed native-import contract. It also freezes the paper's NIST-III/Shorter TCitH tree parameters and exposes the required $\mathbb{F}_{2^{193}}$ backend lift. Version 1.9 implements polynomial-basis arithmetic for that field and a source-pinned eight-element Anemoi permutation as ordinary native rank-one rows. The implementation also exposes a reproducibility blocker rather than hiding it: Blind-UOV reports 240 constraints per level-III permutation, while the public upstream round rule and the published characteristic-two rule give 336 and 360 nonlinear rows, respectively, under the direct closed-Flystel lowering. The full CAP-hash import therefore remains fail-closed. Version 2.0 deliberately selects the aggressive interoperability fork. It freezes an independent profile named *PQ-RBBC-Anemoi-193/336-Sponge-v1*, with four rate and four capacity elements, injective length framing, explicit domain separation, and a 576-bit request-binding

output H_{RBBC} . The native trace constrains input bitness, byte-to-field packing, two permutation calls for the frozen request vector, state chaining, and output decomposition. This removes the 240-row artifact gap as an engineering blocker by giving up bit-exact Blind-UOV compatibility; it does not inherit the paper’s concrete security or 11,644-byte signature claim. At the version-2.0 checkpoint, CAP.Commit and GGM remained open; fork-specific security proof and fresh benchmarking still remain open and production stays fail-closed. Version 2.1 implements a source-grounded direct reference for Protocols 8–10 of Blind-UOV: salted GGM seed derivation, leaf commitments and tapes, random degree-one polynomial coefficients, cross-repetition corrections, the consistency digest, canonical commitment serialization, and the exact byte-level join to H_{RBBC} . A reduced non-secure vector executes all code paths and mutation tests. The production profile has 18 trees, 40,960 leaves, a 5,378-byte hidden commitment, and an exact estimate of 389,974 selected Anemoi permutations, or 131,031,264 nonlinear permutation rows. The production vector and native inter-call row stream have not been materialized, so the parent archive deliberately retains one external assertion and production closure remains false. Version 2.2 lowers the complete reduced CAP fixture and its final $H_{\text{RBBC}}(m, c_r)$ join into one zero-callback native $\mathbb{F}_{2^{193}}$ row stream. The frozen component has 88,282 rows, 59,602 wires, 24 XOF calls, 57 Anemoi permutations, witness-independent topology, and zero external assertions. Input, commitment, and request-hash mutation tests reject stale witnesses. This is an explicitly non-secure 64-bit fixture: it validates the lowering and wire-identity architecture but does not close the 18-tree production profile. Version 2.3 removes the one-rate-block limitation from the native sponge. A frozen 2,450-bit tape trace uses four squeeze blocks, three constrained squeeze-state permutations, 4,858 rows, and 4,845 wires. The same primitive is integrated into an extended CAP fixture in which all eight leaf tapes have the production width. That zero-callback component has 113,802 rows, 85,034 wires, 24 XOF calls, 81 Anemoi permutations, witness-independent topology, and zero external assertions. The extended fixture deliberately retains the 64-bit one-coefficient witness and two tiny trees. It closes only the multi-squeeze engineering blocker; production multi-coefficient polynomial hashing, the 18-tree execution, and fork-specific security qualification remain open. Version 2.4 closes the independent field-arithmetic blocker. A bit-bound $\mathbb{F}_{2^{193}}$ multiplication fixture uses one native multiplication row, and a frozen 2,048-bit, 11-coefficient Horner fixture evaluates at two constrained nonzero distinct points with 20 multiplication rows and zero external assertions. The same generic lowering is integrated with all symbolic extension-mask slices in a deliberately non-secure 386-bit, two-coefficient CAP fixture whose tapes remain 2,450 bits. That combined component has 125,401 rows, 92,816 wires, 14 Horner multiplication rows, witness-independent topology, and zero external assertions. The full 2,048-bit, 18-tree production integration, fork-specific security proof, and fresh benchmarks remain open; the parent archive therefore still retains one external assertion. Version 2.5 closes the next production-shape engineering boundary. One real 2^{11} -leaf tree is executed with the complete 2,048-bit witness, eleven field coefficients, degree-12 extension masks, two consistency points, and 2,450-bit tapes. A bounded-memory lowering hashes 26,126,283 canonical rows and 19,903,324 wire identifiers without retaining the 18,869,935,441-byte expanded stream. The component has zero callbacks and zero external assertions; its peak resident set was 140,932 KiB. This is a non-secure one-tree topology result, not a complete satisfiability archive: the full assignment, the 2^{12} -leaf shard, the other 17 trees, the parent wire join, fork-specific security proof, and fresh proof benchmarks remain open.

1 Status, scope, and non-goals

This is the standalone cryptographic core. It intentionally excludes satellite AKE, radio transport, FLEO relay behavior, handover, revocation distribution, and concrete parameter optimization. Issuance and opening may use large proofs, large public keys, and large private keys. The constrained online path carries only a ticket payload and one compact blind signature.

The construction is designed for the following division of trust.

- The issuing HNCC is *honest but curious*: it authenticates the real identity, enforces a common issuance policy, verifies Π_{issue} , and signs only after acceptance, but later tries to link an observed ticket to an issuance session.

- Holders and network attackers may be malicious and quantum polynomial time.
- Fewer than τ of n opening authorities may be corrupted. Any authorized set of at least τ authorities can open a ticket.
- Issuer signing keys and opening decryption keys are independent even when the organizations operating them overlap.
- Per-user visible metadata is forbidden. Version, epoch, policy, expiration bucket, and key identifiers are common to a sufficiently large anonymity set; otherwise they become an issuer watermark outside the cryptographic game.

Claim boundary. The theorems below are complete reductions for the abstract interfaces and games defined here. They are not a claim that the PQ-RBBC-BUOV-336 fork or threshold Niederreiter papers already implement every required interface. The forked request equations and their visibility are fixed in Section 3.2; the exact added circuit is counted in Section 8.3 and executed in Section 8.4. The resulting proof-size and prover runtime figures remain paper-anchored estimates rather than implementation benchmarks. The executable generator is a relation validator, not a proof backend. The numerical Niederreiter set is a size anchor, not a deployment approval.

2 Notation and post-quantum conventions

The security parameter is λ . All adversaries and reductions are uniform or non-uniform QPT algorithms. A function $\text{negl}(\lambda)$ is negligible. Encodings are canonical, length-prefixed, and domain separated. Concatenation is written $\parallel$. Every hash invocation includes a fixed protocol label and version.

The issuing organization authenticates a real identity rid before the cryptographic issuance protocol. The session identifier sid is used only for replay and quota control and is never copied into a ticket. The common issuance information is

$$\text{ci} = (\text{version}, \text{epoch}, \text{domain}, \text{policy}, \text{expiryBucket}, \text{kid}_{\text{OA}}, \text{kid}_{\text{I}}).$$

Its fixed 32-byte ticket representation is

$$\text{ctx} = \text{SHAKE256}(\text{PQ-RBBC}/\text{CTX} \parallel \text{Encode}(\text{ci}), 256).$$

The federation configuration signature authenticates (ci, ctx) . The digest is computed once outside Π_{issue} and is a public input; the issuer cannot select a per-user value.

For any game G , $\text{Adv}_{\mathcal{A}}^G(\lambda)$ denotes the usual absolute distance from the ideal winning probability. Oracle calls are polynomially bounded.

3 Required primitive interfaces

3.1 Hashing and CAP commitment

We use domain-separated hashes

$$H_{\text{ticket}}, \quad H_{\text{RBBC}}, \quad H_{\text{hold}}, \quad H_{\text{evidence}}.$$

They are modeled as quantum-resistant collision-resistant hashes. A random oracle or QROM instantiation may be used by a concrete component, but the core proof calls only the explicitly named property.

For the version-1.3 circuit audit, fix

$$\begin{aligned} H_{\text{ticket}}(X) &= \text{SHAKE256}(\text{PQ-RBBC}/\text{TICKET} \parallel X, 256), \\ H_{\text{hold}}(k) &= \text{SHAKE256}(\text{PQ-RBBC}/\text{HOLD} \parallel k, 256). \end{aligned}$$

The request-binding hash H_{RBBC} uses the independently frozen PQ-RBBC-Anemoui-193/336 sponge profile. Version 2.1 instantiates the CAP commitment internals as an independent direct reference and canonical byte encoding; its full native row stream is not yet instantiated. The Blind-UOV paper is a framework and size comparator, not a bit-exact implementation claim. This mixed choice retains standardized SHAKE/KMAC for ticket and trace logic while using the native-field sponge only inside the forked blind-request boundary.

There is no separate public commitment to the ticket payload. The accepting proof extracts M , Equation (I2) fixes its digest, and the forked request binds that digest. Publishing an additional c_M would add a primitive and a circuit block without strengthening this chain.

3.2 Forked BUOV request normal form and visibility

From version 2.0 onward, the operational symbol BUOV denotes the independent PQ-RBBC-BUOV-336 fork. It retains the hidden-state request shape of Blind-UOV Protocols 3 and 4, but replaces the unreproduced permutation/hash instance. References to the published Blind-UOV construction are explicit comparisons only. None of its concrete security or size results transfers automatically to this fork.

The blind-signature interface is

$$(\text{bsk}, \text{bvk}) \leftarrow \text{BUOV.KeyGen}(1^\lambda),$$

with an interactive request/sign/unblind protocol and a public verification algorithm

$$\text{BUOV.Vfy}(\text{bvk}, m, \sigma).$$

The core needs more than ordinary blindness: the issuance proof must identify the exact hidden message represented by the request. Protocols 3 and 4 of the 31 October 2025 revision have the user compute [1]

$$c_r \leftarrow \text{CAP.Commit}(r; \rho), \tag{B1}$$

$$y = r + H_{\text{RBBC}}(m, c_r) \in \mathbb{F}_q^o. \tag{B2}$$

Only (y, π_1) is sent to the signer. The values m, r, ρ, c_r are witness/state, not signer-visible request fields. The final signature later contains (c_r, c_x, π_2) . Publishing c_r during issuance would both depart from the paper and give the issuer a direct link to the final signature.

A Blind-UOV-Is target is only 256 bits. Our authorization-binding property is a freely chosen cross-message collision problem, not a fixed-target preimage problem, so its generic quantum scale is only about $2^{256/3} = 2^{85}$ [5, 6]. Version 1.7 instead selects the paper’s Blind-UOV-III/Shorter profile: r and y are 576 bits, the cached public key is 189.2 KB, and the final signature is 11,644 bytes [1]. Version 2.0 keeps the 576-bit interface but treats those key and signature figures as provisional engineering targets, not measurements or theorems for the fork. Define

$$J_m(r, \rho) = r + H_{\text{RBBC}}(m, \text{CAP.Commit}(r; \rho)).$$

The public request is exactly the 72-byte value $\beta = y$; its hidden witness is $b = (r, \rho)$.

Definition 3.1 (PQ-RBBC-BUOV-336 request relation). *For $\beta = y$ and $b = (r, \rho)$, define*

$$R_{\text{req}}^{\text{RBBC-336}}(y; m, r, \rho) = 1 \iff c_r = \text{CAP.Commit}(r; \rho) \wedge y = r + H_{\text{RBBC}}(m, c_r),$$

where c_r is a derived hidden witness. A completed session is intended to return one PQ-RBBC-BUOV-336 fork signature $\sigma = (c_r, c_x, \pi_2)$ on m ; its concrete format and size require fresh validation.

Theorem 3.2 (QROM cross-message request binding). *Let $H_{\text{RBBC}} : \{0, 1\}^* \rightarrow \{0, 1\}^{576}$ be an independent quantum random oracle. Assume the CAP commitment has straightline extraction and a unique committed mask: producing one admissible c_r with two different extracted r values is negligible. If an adversary makes at most q relevant quantum-oracle/evaluation queries, then*

$$\text{Adv}_J^{\text{xmsg}}(\mathcal{A}) \leq \text{Adv}_{\text{CAP}}^{\text{uw}}(\mathcal{B}_1) + \text{Adv}_{\text{CAP}}^{\text{ext}}(\mathcal{B}_2) + O\left(\frac{q^3}{2^{576}}\right).$$

In particular, the generic quantum collision scale is $2^{576/3} = 2^{192}$; at $q = 2^{128}$ the random-function term is $O(2^{-192})$.

Proof. Run the straightline CAP extractor on every admissible commitment relevant to the adversary's output. Abort on extraction failure. If one commitment yields two different masks, output a CAP unique-witness violation. Otherwise each admissible commitment c has a well-defined value $r(c)$ fixed before the domain-separated query $H_{\text{RBBC}}(m, c)$. Quotienting duplicate openings with the same $(c, r(c))$, define

$$Z(m, c) = H_{\text{RBBC}}(m, c) + r(c).$$

For every admissible input, addition of the fixed $r(c)$ is a permutation of the 576-bit oracle range. Thus Z is a pointwise-shifted 576-bit random function. If the adversary gives two accepted openings of one public y for $m \neq m'$, then $(m, c) \neq (m', c')$ and $Z(m, c) = Z(m', c') = y$, a collision. The compressed-oracle collision bound is $O(q^3/2^{576})$ [6]; the stated CAP bad events cover the remaining cases. $\square$

The theorem is an ideal-QROM statement. Version 2.0 supplies a concrete Anemoi-based function with the same 576-bit range, but does not prove that this function realizes an independent quantum random oracle. Fork qualification must therefore include cryptanalysis and a concrete-to-ideal justification or must retain QROM programmability as an explicit heuristic assumption.

Remark 3.3 (Why v1.6 parallel repetition was removed). *Two independently openable 256-bit lanes do not behave as one monolithic 512-bit random function. After fixing m, m' , an attacker can run a BHT claw search for lane 0 and then another for lane 1, for total work about $2 \cdot 2^{85}$ rather than 2^{170} . A larger single request range gives the required collision exponent without adding a correlation assumption that might damage blindness.*

Remark 3.4 (Why ordinary blind signing is insufficient). *If Π_{issue} proves a valid ciphertext but the proof is not tied to the exact blind request β , a malicious holder can prove statement M_0 and ask for a signature on M_1 . The $R_{\text{req}}^{\text{BUOV-III}}$ conjunct is the formal connection that prevents this substitution.*

3.3 Simulation-extractable NIZK argument of knowledge

The proof system $\Pi_{\text{issue}} = (\text{Setup}, \text{Prove}, \text{Vfy})$ is a post-quantum simulation-extractable NIZK argument of knowledge for the issuance relation in Section 4. Its setup produces a CRS; the security proof may use simulation and extraction trapdoors. In the real protocol no trapdoor is available to any party.

The proof may be several megabytes because it is transmitted only during infrequent issuance. This design choice is intentional. Knowledge soundness alone is enough for decoder safety in a simple stand-alone execution; simulation extractability is used to support concurrent sessions and the gated-CCA hybrid proof.

3.4 Compact threshold trace encryption

The threshold encryption interface is

$$\begin{aligned} (\text{tpk}, \text{tsk}_1, \dots, \text{tsk}_n) &\leftarrow \text{TEnc.KeyGen}(1^\lambda, n, \tau), \\ C &\leftarrow \text{TEnc.Enc}(\text{tpk}, \mu; \text{ad}, \rho), \\ d_i &\leftarrow \text{TPDec}(\text{tsk}_i, C), \\ \mu \text{ or } \perp &\leftarrow \text{TComb}(C, \{d_i\}_{i \in I}), \quad |I| \geq \tau. \end{aligned}$$

There is an NP relation

$$R_{\text{enc}}(\text{tpk}, C, \text{ad}; \mu, \rho) = 1 \iff C = \text{TEnc.Enc}(\text{tpk}, \mu; \text{ad}, \rho).$$

Only ciphertexts in the language L_{enc} are promised to be safe inputs to the distributed decoder.

We require the following post-quantum properties.

- (T1) **Correctness on valid inputs.** Combining any τ correct shares for a ciphertext in L_{enc} returns its plaintext.
- (T2) **τ -privacy under chosen plaintext.** A QPT adversary holding fewer than τ secret shares cannot distinguish encryptions of two equal-length plaintexts.
- (T3) **Unique opening.** It is infeasible to find $C, \mu, \rho, \mu', \rho'$ with $\mu \neq \mu'$ such that both $R_{\text{enc}}(\text{tpk}, C, \text{ad}; \mu, \rho) = 1$ and $R_{\text{enc}}(\text{tpk}, C, \text{ad}; \mu', \rho') = 1$.
- (T4) **Robust shares.** Invalid or inconsistent partial decryptions are detected by TComb or by a component proof/MAC before a plaintext is accepted.

Ordinary IND-CCA security of the bare encryption is not required and is not claimed.

3.5 Opening authorization

An independent post-quantum authorization mechanism verifies

$$\text{GateAuthVfy}(\text{gavk}, d_M, \text{caseID}, H_{\text{evidence}}(E), \text{auth}) = 1,$$

where $d_M = H_{\text{ticket}}(\text{Encode}(M))$. The authorization binds a ticket digest, case identifier, evidence digest, expiry, and purpose. Its unforgeability and policy governance are orthogonal to trace-ciphertext privacy, but the binding is required to prevent using an authorization for a different ticket.

4 Ticket format and issuance relation

The hidden ticket payload is

$$M = (\text{ctx}, \text{sn}, h, C), \quad h = H_{\text{hold}}(k_{\text{hold}}), \quad C = \text{Ntr.Enc}(\text{tpk}, \text{ctx}, h; \text{rid}, \text{sn}, e).$$

For the v1.3 profile all fields are fixed width and

$$\text{Encode}(M) = \text{ctx} \parallel \text{sn} \parallel h \parallel s \parallel c_\mu \parallel \gamma \in \{0, 1\}^{2944},$$

namely $32 + 16 + 32 + 208 + 48 + 32 = 368$ bytes. No variable-length field or user-selected policy string occurs in the payload. The serial number $\text{sn} \leftarrow \{0, 1\}^{128}$ and holder key $k_{\text{hold}} \leftarrow \{0, 1\}^{256}$ are fresh. The serial appears both outside and inside the trace ciphertext so that an opener can detect payload splicing. The holder secret itself never appears in the ticket. Define

$$m = H_{\text{ticket}}(\text{Encode}(M)).$$

The public issuance statement and private witness are

$$\begin{aligned} x &= (\text{pp}, \text{ctx}, \text{sid}, \text{rid}, \beta), \\ w &= (M, r, \rho, k_{\text{hold}}, e). \end{aligned}$$

The relation $R_{\text{issue}}(x; w) = 1$ iff all of the following hold:

$$\text{TicketShape}(M, \text{ctx}) = 1, \quad (\text{I1})$$

$$m = H_{\text{ticket}}(\text{Encode}(M)), \quad (\text{I2})$$

$$R_{\text{req}}^{\text{BUOV-III}}(\beta; m, r, \rho) = 1, \quad (\text{I3})$$

$$M.h = H_{\text{hold}}(k_{\text{hold}}), \quad (\text{I4})$$

$$R_{\text{enc}}(\text{tpk}, M.C, \text{Encode}(\text{ctx}, M.\text{sn}, M.h); \text{rid} \| M.\text{sn}, e) = 1. \quad (\text{I5})$$

The real identity and session identifier are public to the issuer during issuance but are not part of M . Equation (I5) binds the hidden trace plaintext to the identity authenticated in that exact session.

5 Construction

5.1 Setup

$\text{Core.Setup}(1^\lambda, n, \tau)$ performs:

1. generate Π_{issue} parameters and the domain-separated hash parameters;
2. generate issuer keys $(\text{bsk}, \text{bvk}) \leftarrow \text{BUOV.KeyGen}(1^\lambda)$;
3. generate independent opening keys $(\text{tpk}, \text{tsk}_1, \dots, \text{tsk}_n) \leftarrow \text{TEnc.KeyGen}(1^\lambda, n, \tau)$;
4. publish (ci, ctx) authenticated by a federation configuration signature.

No party is allowed to derive bsk and any tsk_i from the same seed or key hierarchy.

5.2 Blind relation-bound issuance

For authenticated identity rid and fresh issuer-side sid :

1. The holder samples $\text{sn}, k_{\text{hold}}, e$ and computes

$$h = H_{\text{hold}}(k_{\text{hold}}), \quad C = \text{Ntr.Enc}(\text{tpk}, \text{ctx}, h; \text{rid}, \text{sn}, e)$$

according to Equations (N1)–(N7), constructs M , and computes $m = H_{\text{ticket}}(\text{Encode}(M))$.

2. It samples $r \in \{0, 1\}^{576}$ and ρ , computes

$$c_r = \text{CAP.Commit}(r; \rho), \quad y = r + H_{\text{RBBC}}(m, c_r),$$

sets the public request $\beta = y$, and produces

$$\pi_{\text{issue}} \leftarrow \Pi_{\text{issue}}.\text{Prove}(\text{crs}, x, w).$$

3. The issuer checks authenticated identity, epoch, quota, replay status, and

$$\Pi_{\text{issue}}.\text{Vfy}(\text{crs}, x, \pi_{\text{issue}}) = 1.$$

Only then does it execute $z \leftarrow \text{BUOV.Respond}(\text{bsk}, y)$.

4. The holder computes

$$\sigma \leftarrow \text{BUOV.Finalize}(m, c_r, y, z; r, \rho)$$

and accepts only if $\text{BUOV.Vfy}(\text{bvk}, m, \sigma) = 1$.

The proof π_{issue} is not an extra proof placed beside the fork’s well-built-request proof. It is the strengthened replacement for that proof: the original request equations are retained and the ticket, identity, holder, and trace-ciphertext conjuncts are appended to the proved relation. Thus issuance sends one augmented relation proof.

The spendable ticket is $T = (M, \sigma)$. Neither β , π_{issue} , rid , nor sid is transmitted with T .

5.3 Ticket verification

$\text{Core.VerifyTicket}(T)$ recomputes $m = H_{\text{ticket}}(\text{Encode}(M))$, checks canonical payload encoding and the public common-information policy, then returns $\text{BUOV.Vfy}(\text{bvk}, m, \sigma)$.

This check proves neither public ciphertext validity nor identity correctness by itself. Those facts follow from the certified-issuance theorem and the trust assumption that the issuer signs only after accepting Π_{issue} .

5.4 Signature-gated threshold opening

For an opening request

$$Q = (T, E, \text{caseID}, \text{auth}),$$

each honest opening authority runs the following gate before touching its decoder:

1. require $\text{Core.VerifyTicket}(T) = 1$;
2. require the common epoch and key identifiers to be acceptable;
3. compute $d_M = H_{\text{ticket}}(\text{Encode}(M))$ and require a valid, unused, purpose-limited authorization bound to $(d_M, \text{caseID}, H_{\text{evidence}}(E))$;
4. only after all checks pass, output $d_i = \text{TPDec}(\text{tsk}_i, M.C)$, bound to the same request digest.

The combiner accepts a reconstructed plaintext $\text{rid}' \parallel \text{sn}'$ only if robust share verification succeeds and $\text{sn}' = M.\text{sn}$. Otherwise it outputs $\perp$.

Critical API rule. The implementation MUST NOT export $\text{TPDec}(\text{tsk}_i, C)$ as a callable production endpoint. The only exposed operation is $\text{Core.OpenShare}(\text{tsk}_i, Q)$, which internally performs the gate. This is a security boundary, not an application convention.

6 Security games

6.1 Certified-language soundness

The challenger maintains a log $\mathcal{L}$ of accepted issuance sessions. In the analysis it uses the extractor to associate each completed session with (rid, M, w) . An adversary wins **CertSound** if it outputs a ticket $T = (M, \sigma)$ for which $\text{Core.VerifyTicket}(T) = 1$ and either:

- (a) M is not associated with any completed accepted issuance in $\mathcal{L}$; or
- (b) every associated extraction fails one of Equations (I1)–(I5).

Distinct messages are counted by their canonical ticket digest. This game captures both a new signature and a proof/request substitution.

6.2 Issuer unlinkability

An honest-protocol curious issuer chooses or observes two authenticated identities $\text{rid}_0, \text{rid}_1$ that receive the same ci . Two honest holders execute issuance concurrently. The issuer obtains its full protocol views, after which it receives the two resulting tickets in random order and guesses the order. It may retain all issuance state but receives no opening shares. The advantage over $1/2$ is issuer-unlinkability advantage.

This game allows the issuer to know which identity participated in each issuance; it hides which later ticket came from which session. It cannot hide timing, a uniquely sized policy, or a per-user expiration bucket, which is why common metadata is a stated system requirement.

6.3 Gated chosen-ciphertext trace privacy

The game **t-GCCA** is parameterized by threshold τ . The adversary receives all public parameters and fewer than τ opening secret shares. It may run malicious-user issuance sessions for identities it is authorized to use, receiving completed tickets, and may submit opening queries for any validly authorized ticket other than the exact challenge ticket digest.

For the challenge it selects equal-length identities $\text{rid}_0, \text{rid}_1$ and common ticket metadata. The challenger samples $b \leftarrow \{0, 1\}$, forms a valid ticket whose trace ciphertext encrypts $\text{rid}_b \parallel \text{sn}^*$, and returns $T^* = (M^*, \sigma^*)$. The adversary wins by guessing b .

The HNCC issuer is not corrupted in this game because it necessarily observes the authenticated identity at issuance. HNCC privacy is instead covered by issuer unlinkability. The game models an outside observer, an evidence processor, and fewer than τ opening authorities.

Remark 6.1 (Why the game is not ordinary CCA). *The opening oracle accepts only issuer-certified ticket payloads, not arbitrary ciphertexts. Calling the result IND-CCA for bare TEnc would be incorrect. The security statement is protocol-specific: certification plus the gate turns chosen-ciphertext access into chosen-certified-ciphertext access.*

6.4 Trace soundness and non-frameability

An adversary wins **TraceSound** if a valid ticket produced from a completed accepted issuance authenticated as rid opens to $\text{rid}' \neq \text{rid}$, or if it opens with an external serial number different from the encrypted serial. The issuer is honest in following the protocol. The holder may be malicious.

7 Security results

7.1 Correctness

Theorem 7.1 (End-to-end correctness). *For an honest issuance authenticated as rid , an honestly produced ticket $T = (M, \sigma)$ verifies. Any authorized set of at least τ honest opening authorities reconstructs $\text{rid} \parallel M.\text{sn}$, except with the sum of the component correctness errors.*

Proof. Honest computation satisfies every conjunct of R_{issue} , so Π_{issue} accepts. Blind-signature correctness gives a verifying signature on $H_{\text{ticket}}(\text{Encode}(M))$. Equation (I5) gives a valid encryption of $\text{rid} \parallel M.\text{sn}$. Threshold-encryption correctness on valid inputs reconstructs that plaintext from any τ correct shares. The serial cross-check therefore succeeds. $\square$

7.2 Blind-signing message binding

Lemma 7.2 (Request uniqueness). *Let $\beta = y$. If the PQ-RBCC-BUOV-336 request relation accepts valid openings for both m and $m' \neq m$, then those values constitute a winning output in the cross-message game of Theorem 3.2.*

Proof. Expanding the two accepted relations gives distinct inputs (m, c_r) and (m', c'_r) whose shifted-oracle values are both y . This is precisely the cross-message collision covered by Theorem 3.2, including the CAP extraction and unique-witness bad events. $\square$

Lemma 7.3 (Request/proof binding). *Suppose an issuance proof for β is accepting and extractable. Except with cross-message request-collision probability, the message signed through β is exactly $H_{\text{ticket}}(\text{Encode}(M))$ for the payload M in the extracted witness. A different payload cannot verify under the same signed digest except by finding a collision in H_{ticket} .*

Proof. Extraction yields $(M, r, \rho, \dots)$ satisfying Equations (I2)–(I3). Equation (I2) fixes m to the canonical ticket digest. Lemma 7.2 makes the message represented by β equal to this m . If a different canonical payload $M' \neq M$ verifies under that signed value, then $H_{\text{ticket}}(\text{Encode}(M')) = H_{\text{ticket}}(\text{Encode}(M))$, a collision. $\square$

Theorem 7.4 (Certified-language soundness). *For any QPT adversary $\mathcal{A}$, there exist QPT reductions $\mathcal{B}_1, \dots, \mathcal{B}_4$ such that*

$$\text{Adv}_{\text{Core}, \mathcal{A}}^{\text{CertSound}} \leq \text{Adv}_{\Pi_{\text{issue}}, \mathcal{B}_1}^{\text{SE-AoK}} + \text{Adv}_{\text{BUOV}, \mathcal{B}_2}^{1\text{mUF}} + \text{Adv}_{J, \mathcal{B}_3}^{\text{xmsg}} + \text{Adv}_{H_{\text{ticket}}, \mathcal{B}_4}^{\text{coll}}.$$

By Theorem 3.2, the third term is at most the two CAP bad-event advantages plus $O(q^3/2^{576})$. Consequently, every verifying ticket is associated with an accepted issuance witness satisfying R_{issue} , except with negligible probability.

Proof. Consider a verifying output (M, σ) . If its canonical digest was not the message of a completed blind-signing session, then the output is a one-more forgery. Otherwise consider the corresponding accepting issuance proof. Simulation extraction returns a witness or violates the SE-AoK property. Lemma 7.3 associates the signed digest with the extracted payload, except under the remaining claw and collision bad events. The extracted witness then satisfies every issuance conjunct, including ciphertext validity and identity binding. $\square$

Corollary 7.5 (Certified ciphertext validity). *If $\text{Core.VerifyTicket}(M, \sigma) = 1$, then $M.C \in L_{\text{enc}}$ and it has an opening to the authenticated issuance identity and the external serial number, except with the bound in Theorem 7.4.*

7.3 Decoder safety

Theorem 7.6 (Signature-gated decoder safety). *Assume honest opening authorities expose only Core.OpenShare and execute its checks in the specified order. The probability that an honest authority evaluates TPDec on a ciphertext outside L_{enc} is at most*

$$\text{Adv}_{\text{Core}}^{\text{CertSound}} + \text{Adv}_{\text{GateAuth}}^{\text{forge}} + \text{Adv}_{H_{\text{ticket}}}^{\text{coll}}.$$

Proof. Reaching TPDec implies that the ticket signature and the opening authorization both verified for the same ticket digest. By Corollary 7.5, a verifying ticket contains a ciphertext in L_{enc} . A mismatch between the ticket checked by the gate and the ciphertext passed to the decoder either violates the atomic API invariant or requires a digest collision. A request not authorized for that digest is an authorization forgery. Taking a union bound gives the result. $\square$

Remark 7.7. *This theorem explains what is gained by moving the validity proof out of the online ticket. The large issuance proof is amortized, while the small issuer signature becomes a certificate that the ciphertext was once proved valid. The theorem fails immediately if an implementation offers a raw partial-decryption endpoint.*

7.4 Issuer unlinkability

Theorem 7.8 (Issuer unlinkability). *For an honest-protocol curious issuer, common visible metadata, and two honest holders, there exist QPT reductions such that*

$$\text{Adv}_{\text{Core}}^{\text{unlink}} \leq \text{Adv}_{\text{BUOV}}^{\text{blind}} + 2\text{Adv}_{\Pi_{\text{issue}}}^{\text{zk}} + \text{Adv}_{\text{TEnc}}^{\text{t-IND-CPA}} + \text{Adv}_{H_{\text{ticket}}}^{\text{coll}}.$$

Proof. First replace each real issuance proof with a simulated proof; zero knowledge bounds the change. The remaining request/response views are ordinary fork views, so the assumed signer-blindness property permits swapping the two hidden messages without the issuer detecting the swap. Threshold-encryption privacy prevents the issuer, which holds no opening threshold, from matching a final trace ciphertext to either known issuance identity. Canonical digest collisions are excluded. After these replacements, the issuer's view is independent of the final permutation, so its success probability is $1/2$. $\square$

Operational qualifier. The theorem is cryptographic, not traffic-analytic. Issuance batching, fixed message sizes, delayed activation, and common metadata are required to keep timing and formatting from becoming a link tag.

7.5 Trace correctness and non-frameability

Theorem 7.9 (Trace soundness). *Under certified-language soundness and unique opening of TEnc, a malicious holder completing issuance while authenticated as rid cannot produce a verifying ticket that opens to $\text{rid}' \neq \text{rid}$ or to a different serial number, except with negligible probability.*

Proof. By Theorem 7.4, a verifying ticket is associated with an extracted accepted issuance witness. Equation (I5) states that its ciphertext opens to $\text{rid} \parallel M.\text{sn}$. If threshold opening returns a different plaintext while all robust-share checks pass, either threshold correctness fails or the ciphertext has two different valid openings. The latter violates unique opening. The explicit serial comparison catches any external/internal serial mismatch. $\square$

Remark 7.10 (Issuer framing). *The theorem assumes an honest-protocol issuer because the issuer learns rid and controls the signing key. A malicious issuer could directly sign a payload encrypting an arbitrary victim identity. Removing that trust requires a threshold blind issuer, an auditable issuance log with privacy-preserving accountability, or retaining a publicly checked proof. It is not solved by the present compact-ticket core.*

7.6 Gated chosen-ciphertext trace privacy

Theorem 7.11 (τ -GCCA trace privacy). *Assume fewer than τ opening authorities are corrupted. For every QPT adversary $\mathcal{A}$ in the game of Section 6.3, there exist QPT reductions $\mathcal{B}_i$ such that*

$$\begin{aligned} \text{Adv}_{\text{Core}, \mathcal{A}}^{\text{t-GCCA}} &\leq \text{Adv}_{\text{TEnc}, \mathcal{B}_0}^{\text{t-IND-CPA}} + \text{Adv}_{\Pi_{\text{issue}}, \mathcal{B}_1}^{\text{SE-AoK}} + \text{Adv}_{\text{BUOV}, \mathcal{B}_2}^{\text{1mUF}} \\ &\quad + \text{Adv}_{\text{TEnc}, \mathcal{B}_3}^{\text{unique}} + \text{Adv}_{J, \mathcal{B}_4}^{\text{xmsg}} \\ &\quad + \text{Adv}_{H_{\text{ticket}}, \mathcal{B}_5}^{\text{coll}}. \end{aligned}$$

Proof. We use a sequence of games.

Game 0. This is the real t-GCCA game.

Game 1: extract accepted issuance sessions. For every accepted adversarial issuance proof, use the SE-AoK extractor to record

$$(d_M, M, \text{rid}, e, k_{\text{hold}}, r, \rho, \dots)$$

in a private table $\mathcal{L}$. Abort if an accepting proof has no valid witness. The difference from Game 0 is bounded by SE-AoK security. Request binding and hash collision resistance ensure that the final signed digest is the digest recorded in $\mathcal{L}$.

Game 2: reject unlogged verifying tickets. If the adversary submits a verifying ticket whose digest is neither the challenge digest nor in $\mathcal{L}$, abort. Such a ticket supplies an additional valid message/signature pair not obtained from a completed issuance, which yields a one-more forgery. Proof/request substitution and digest ambiguity are charged to the corresponding terms in the theorem.

Game 3: simulate the opening oracle from the issuance table. For every non-challenge verifying ticket, look up the extracted plaintext in $\mathcal{L}$ and return it after the authorization checks. This has the same result as threshold opening by correctness and unique opening, but it uses no honest opening secret share and never invokes a decryption oracle. A ticket that reuses the challenge ciphertext under a newly certified digest requires an accepted proof containing a valid opening of that ciphertext. Extraction then reveals the challenge plaintext. If the extracted opening disagrees with the challenge plaintext, it violates unique opening; if it agrees, the reduction can immediately guess the threshold-encryption challenge bit. Thus challenge re-certification does not create an unanswerable decryption query.

Game 4: replace the challenge encryption. Embed the τ -IND-CPA challenge of TEnc as $M^*.C$. The reduction is given fewer than τ shares exactly as in the base game. It samples the remaining visible ticket fields identically. Because it generated the issuer key and knows the complete challenge message m^* , it locally executes the assumed fork protocol and obtains one exactly distributed final signature; it does not need the unknown trace-encryption randomness. The challenge issuance transcript is not given to the adversary in this game, and π_{issue} influences only the accept/reject gate, not the distribution of the response after acceptance. Therefore the reduction may internally bypass that gate when manufacturing the challenge ticket without changing any value visible to the adversary. All opening queries are answered from $\mathcal{L}$ as in Game 3. Therefore changing the encrypted identity from rid_0 to rid_1 changes the adversary's view by at most the τ -IND-CPA advantage of TEnc.

In the final game the challenge bit is independent of the adversary's view. Summing the hybrid differences proves the theorem. $\square$

Why the reduction is useful. The large issuance proof is not merely a validity check. Its extractor gives the simulator the plaintext and randomness of every non-challenge ciphertext that can ever obtain a valid issuer signature. Therefore the simulator can answer gated opening queries without a base decryption oracle. This is the precise reason a CPA-secure compact threshold encryption can suffice inside the gated protocol even though it would not suffice under an unrestricted CCA API.

7.7 One-more ticket unforgeability

Theorem 7.12 (Ticket one-more unforgeability). *After q completed accepted blind-issuance sessions, producing more than q distinct verifying ticket digests has probability at most the one-more-unforgeability advantage of the selected fork, plus the cross-message request-collision and ticket-hash collision terms.*

Proof. Map every completed issuance to the unique signed digest using Lemma 7.3. More distinct verifying digests than completed sessions then give more valid message/signature pairs than signing responses, unless two payloads share a digest or Lemma 7.2 fails. $\square$

8 Candidate concrete instantiations

8.1 Augmented well-built-request compiler for the BUOV-336 fork

The Blind-UOV framework already requires the user to prove that its masked request is well built. We strengthen that proved language instead of adding a separate policy proof. Let

$$R_{\text{wb}}(y; m, r, \rho) = 1$$

denote Equations (B1)–(B2). The native well-built-request language is replaced by one augmented language:

$$R_{\text{aug}}(\text{pp}, \text{ctx}, \text{sid}, \text{rid}, y; M, r, \rho, k_{\text{hold}}, e) = 1 \iff R_{\text{issue}}(x; w) = 1,$$

where $\beta = y$ and Equation (I3) is R_{wb} . The issuer executes the forked signer response only after verifying a post-quantum SE-NIZK argument for R_{aug} .

Compiler interface. The integration replaces the native verification hook:

$$\text{VerifyWellBuilt}(y, \pi_1) \longrightarrow \Pi_{\text{issue}}.\text{Vfy}(\text{pp}, \text{ctx}, \text{sid}, \text{rid}, y, \pi_{\text{issue}}).$$

The response and finalization shapes are intended to follow the source framework. The spendable object is a PQ-RBBC-BUOV-336 fork signature; calling it a genuine published Blind-UOV-III signature is forbidden until interoperability is restored, which is not the selected version-2.0 goal.

Concrete circuit boundary. The implementation SHOULD expose the following fixed interface to the proof compiler:

$$\begin{aligned} \text{pub}_{\text{aug}} &= (\text{ctx}, \text{sid}, \text{rid}, \text{tpk}, \text{bvk}, y), \\ \text{wit}_{\text{aug}} &= (\text{sn}, k_{\text{hold}}, e, r, \rho). \end{aligned}$$

The circuit derives C , h , M , and m internally rather than accepting duplicate witness copies. Its constraint system is the conjunction

$$\mathcal{C}_{\text{aug}} = \mathcal{C}_{\text{shape}} \wedge \mathcal{C}_{\text{ticketHash}} \wedge \mathcal{C}_{\text{RBBCmask}} \wedge \mathcal{C}_{\text{holder}} \wedge \mathcal{C}_{\text{trace}},$$

where

$$\begin{aligned} \mathcal{C}_{\text{shape}} &: \text{TicketShape}(M, \text{ctx}) = 1, \\ \mathcal{C}_{\text{ticketHash}} &: m = H_{\text{ticket}}(\text{Encode}(M)), \\ \mathcal{C}_{\text{RBBCmask}} &: c_r = \text{CAP.Commit}(r; \rho) \wedge y = r + H_{\text{RBBC}}(m, c_r), \\ \mathcal{C}_{\text{holder}} &: h = H_{\text{hold}}(k_{\text{hold}}), \\ \mathcal{C}_{\text{trace}} &: R_{\text{Ntr}}(\text{tpk}, C, \text{ctx}, h; \text{rid}, \text{sn}, e) = 1. \end{aligned}$$

The proof statement includes `sid` even though it is absent from the ticket. Consequently, replaying the same proof in a new issuance session changes the statement and fails verification, while the final credential gains no session link tag.

Theorem 8.1 (Conditional security inheritance of the augmented issuance proof). *Assume the complete PQ-RBBC-BUOV-336 fork, with its selected CAP and H_{RBBC} profile, is post-quantum one-more unforgeable and honest-protocol signer blind for its native well-built-request language. Assume also that Π_{issue} is post-quantum zero knowledge and simulation-extractable for R_{aug} . Replacing the fork’s native well-built-request proof by π_{issue} preserves those two assumed properties, up to the SE-NIZK, cross-message request, and hash-collision terms already shown in Theorems 7.4 and 7.8.*

Proof. For one-more unforgeability, a forger against the augmented scheme is also a forger against the completed fork with its native well-built proof: the reduction verifies the augmented proof and forwards each accepted y to the fork signer instance. If that interface requires a native well-built proof, the reduction extracts (m, r, ρ) and generates it. Extraction failure is charged to SE-AoK. The augmented relation restricts the fork’s native well-built language.

For blindness, replace the augmented proof by a simulated proof. The remaining transcripts are ordinary fork transcripts, so the assumed fork blindness game applies directly. The public authenticated identity is session information but is absent from the final ticket; threshold-encryption privacy hides it inside $M.C$. Any attempt to make the same masked request represent a different ticket digest triggers Lemma 7.2, and any two payloads with one ticket digest trigger a collision in H_{ticket} . $\square$

This theorem is no longer a reduction to the published Blind-UOV parameter set. Its first premise is a version-2.0 proof obligation. The executable sponge and row tests establish functional consistency only; they do not establish one-more unforgeability or blindness of the complete fork.

Corrected concretization in version 1.7. The request and witness are now fixed:

$$\begin{aligned} \beta &= y, & b &= (r, \rho), \\ m &= H_{\text{ticket}}(\text{Encode}(M)), & c_r &= \text{CAP.Commit}(r; \rho), \\ y &= r + H_{\text{RBBC}}(m, c_r). \end{aligned}$$

The signer sees only the 72-byte value y , not c_r or m . Earlier drafts incorrectly exposed c_r and claimed that CAP binding plus hash collision resistance implied request uniqueness. That argument fails when an attacker chooses different commitments on both sides. Version 1.6 restored the paper ABI but then incorrectly multiplied the security of two independently openable NIST-I lanes. Version 1.7 instead uses a single 576-bit Blind-UOV-III target and Theorem 3.2. The removed ticket commitment c_M remains unnecessary: extraction supplies M , while ticket-hash collision resistance prevents a second payload under one digest.

Executable refinement through version 2.4. Introduce the hidden 576-bit value

$$h_{\text{RBBC}} = H_{\text{RBBC}}(m, c_r).$$

The executable incremental circuit allocates r and h_{RBBC} as secret binary inputs, enforces their bitness, and materializes all 576 linear equations $y_j + r_j + h_{\text{RBBC},j} = 0$. Its only remaining external assertion is the narrower native subrelation

$$h_{\text{RBBC}} = H_{\text{RBBC}}(m, \text{CAP.Commit}(r; \rho)),$$

where m is passed directly from the circuit-produced ticket-digest wires. Thus a future native importer cannot choose a different y , r , hash image, or ticket message without failing an ordinary circuit row. Version 2.0 replaced the test hash-image computation with the forked Anemoi sponge while retaining a test-only CAP commitment. Version 2.1 adds a strict ABI that accepts only the canonical 5,378-byte production-profile CAP encoding and its frozen profile fingerprint before

computing H_{RBBC} . This closes the byte-level join; the full native row identities across the 18 CAP trees and the hash calls remain external. Version 2.2 closes those identities for the reduced fixture. Version 2.3 also closes every squeeze-state identity for production-width 2,450-bit leaf tapes inside a deliberately small extended fixture. The production parent relation still keeps one external assertion. Version 2.4 natively closes the generic 2,048-bit Horner gadget and exercises its symbolic-mask integration in a two-coefficient CAP fixture; the full 11-coefficient, 18-tree composition is not yet materialized. Version 2.5 executes the full eleven-coefficient path and every degree-12 symbolic mask slice in one real 2,048-leaf production tree shape, while deliberately stopping short of a full assignment archive and the 18-tree parent composition.

Frozen Blind-UOV comparison profile. Version 1.7 fixes the 31 October 2025 revision and its *Blind-UOV-III*, *Shorter*, *TCitH* row. The paper gives security parameter $\lambda = 192$, $\bar{n}_r = 576$ bits, $\bar{n}_x = 1472$ bits, a 189.2 KB public key, and an 11,644-byte final signature [1]. Version 2.4 retains the 576-bit request width, but the key and signature figures are comparison targets only. They must be regenerated for the independent fork before any online-size claim is labelled measured or security-qualified.

Correct backend separation. Blind-UOV has two different proof objects. Its CAP commitment and final CAP proof are carried by the final blind signature. Before the signer responds, the user separately sends a NIZK π_1 proving that the masked request and CAP commitment are well formed. Our π_{issue} replaces and augments this π_1 ; it does *not* replace the final CAP proof. The augmented proof still replaces only π_1 , not the final CAP proof. Nevertheless, version 2.4 may not assert that the final signature remains 11,644 bytes: the extra forked permutation work and any changed transcript must be implemented and benchmarked first.

Native import contract and field boundary. For NIST-III/Shorter/TCitH, Table 2 of the paper fixes $\tau = 18$: two 2^{12} -leaf trees and sixteen 2^{11} -leaf trees, $T_{\text{open}} = 174$, an explicit proof-of-work of 9 bits, and total proof-of-work 13.9 bits. The SNARK-friendly level-III profile uses an eight-element Anemoi state over $\mathbb{F}_{2^{193}}$, with 240 R1CS constraints per permutation [1]. In particular, the abstract ρ contains the CAP salt and GGM-tree randomness; it is not the 32-byte nonce used by the test adapter.

The paper specifies the algorithms but does not ship an executable CAP constraint generator. Version 2.5 uses `pq_rbbc_native_profile.py` as the fail-closed contract for the fork. Closure requires a source and parameter digest, a deterministic native row-stream digest, zero external assertions, witness-independent topology, verified fork vectors, message/mask/hash-image wire identity, and four independent tamper rejections. It additionally requires all seven named CAP/GGM/transcript components, a fork-specific security proof, and fresh signature-size benchmarks. Claiming bit-exact Blind-UOV compatibility is an explicit audit failure. The complete augmented relation must also be lifted from the portable $\mathbb{F}_2$ archive into $\mathbb{F}_{2^{193}}$. This structural certificate prevents accidental under-constraint; it is not a substitute for cryptographic review. The official Anemoi repository supplies permutation and equation-generation code, but not a complete CAP importer [2].

Source-pinned native Anemoi component in version 1.9. The companion implementation `pq_rbbc_anemoi_f193.py` uses the polynomial basis defined by

$$X^{193} + X^8 + X^7 + X^6 + X^5 + X^4 + X^2 + X + 1,$$

the Conway polynomial $f_{2,193}$ published by Luebeck [4]. It pins the official Anemoi repository at commit `3e86ff0cafa54839709b2fa2de0e75d7dd2db464`, together with the SHA-256 digest of its Sage source. For security level 192, four columns, and $\alpha = 3$, that source derives 14 rounds. The explicit characteristic-two estimate in the Anemoi paper instead derives 15 rounds [3].

The characteristic-two closed-Flystel equations contain two cubic terms. The implemented direct lowering uses six multiplication rows per column per round: four square/cube rows and two closed equations. Consequently the pinned 14-round permutation has $14 \cdot 4 \cdot 6 = 336$ nonlinear rows; the paper-rule 15-round variant would have 360. Blind-UOV reports 240 constraints per level-III permutation but publishes neither the bit-exact parameter file nor a constraint stream reproducing that number [1]. These values may reflect different unpublished parameters or an optimized lowering, but the public artifacts do not decide which. Version 1.9 therefore records the 240/336/360 gap as unresolved and forbids production closure.

For the frozen zero-state topology, the native component archive contains 336 nonlinear rows, eight final-state binding rows, and 352 allocated wires. Its deterministic row-stream digest is `25deba5f7fa3f54f1ccc2fd165f2755f8d7137eaa924def12f0c28ba5cdbae4d`. Regression tests check field reduction and inversion, parameter derivation, direct-versus-constrained outputs on three inputs, witness-independent row topology, and stale-witness rejection after input or output mutation. This is one permutation component only: it is not a sponge, CAP.Commit, GGM tree, Fiat-Shamir transcript, or complete Blind-UOV hash relation. Independent Sage-generated vectors and all seven CAP components are still required.

Independent sponge and request hash in version 2.0. The selected profile is *PQ-RBBC-Anemoi-193/336-Sponge-v1*. Its fingerprint is

`4fa0eb276ebba70a9f6c2f38f3f55d19
7c094121a2b614cc6ef9b7e8522cac87.`

It divides the eight-element state into four rate and four capacity elements, so both rate and capacity are 772 bits. For domain string D and payload P , the absorbed bit string is the injective encoding

$$\text{PQRBBC-SPONGE-V1} \parallel \text{u16le}(|D|) \parallel D \parallel \text{u64le}(|P|) \parallel P \parallel \text{pad}_{10*1},$$

with bytes and polynomial-basis coefficients interpreted least-significant bit first. The request hash uses domain `PQ-RBBC/v2.0/H_RBBC`; its payload is an injective ordered-tuple encoding of (m, c_r) , and its output is the first 576 squeezed bits.

For the frozen 32-byte message and 48-byte test-commitment vector, the encoded payload is 118 bytes and absorption takes two permutations. The native trace contains 944 input-bitness rows, 672 permutation multiplication rows, 579 output-bitness rows, and 43 linear packing/chaining rows: 2,238 rows and 2,235 wires in total. Its deterministic row-stream SHA-256 is

`3aa60bc6d8d507003fb541a6ac991e88
da58aea05b7b278ab7e1859772aac9ed.`

Tests reject payload and output mutation, compare constrained and direct hashes on four payloads, check domain and length separation, and prove that the row topology is witness-independent for fixed public lengths.

This closes the permutation-to-hash engineering layer. The historical 48-byte commitment remains only a test fixture for the incremental circuit; it is not the production CAP encoding.

Source-grounded CAP.Commit reference in version 2.1. The companion module `pq_rbbc_cap_commit.py` directly instantiates the initial-commitment logic of Protocols 8–10 of the cited Blind-UOV revision. It implements salted GGM seed derivation, leaf-seed commitments, leaf-tape expansion, degree-one polynomial coefficients, corrections forcing a common constant term across repetitions, polynomial-universal consistency hashing, mask derivation, and later signature-append deltas. Because the paper does not provide an executable CAP generator, this is an independently named fork rather than a bit-exact reproduction. Its relation identifier is `pq-rbbc/cap/tcith-iii/anemoi-193-336/v1` and its profile fingerprint is

2ac471f8d7c6cb4e6352bbc5a2eb7f93
94b807ff132aec8cadebd696f7b1fa38.

The production shape commits a 2,048-bit upper-bound witness consisting of a 576-bit mask and 1,472 later signature bits. It uses two 2^{12} -leaf trees and sixteen 2^{11} -leaf trees. Protocol 8 needs an injection of all leaves into nonzero extension-field elements. For a power-of-two leaf count, $\mathbb{F}_{2^k}^*$ has only $2^k - 1$ elements; this fork therefore uses the safe degrees $\mu = 13$ and $\mu = 12$, rather than 12 and 11. It fixes $\rho = 16$, two $\mathbb{F}_{2^{193}}$ consistency evaluation points (386 bits), and a 2,450-bit tape per leaf. These are fork parameters and require their own security analysis.

The canonical commitment encoding is exactly 5,378 bytes. It contains the profile fingerprint, two salt elements, the 386-bit h_2 , a 386-bit consistency value, and 17 pairs of 2,048-bit and 386-bit corrections. It is hidden inside π_{issue} : the signer still receives exactly the 72-byte masked target y , not c_r or m . A strict API rejects reduced fixtures, wrong profile fingerprints, or any non-canonical length before joining these bytes to $H_{\text{RBBC}}(m, c_r)$.

The reduced, explicitly non-secure two-tree fixture freezes profile fingerprint

5d067a55e2ea9104b2604dc7efa393f4
4d1ce1880c3974bdcaae32aeb825f2ea

and commitment digest

07a09a4f623233586af7ebca90d0eeba
7d6a5bb94ff86c7dff29ade7be79b800.

It executes 23 domain-separated XOF calls. Regression tests verify its frozen output, correction identities, mask/signature round trips, domain separation, the strict H_{RBBC} join, and rejection after salt, root, message, profile, or length mutation.

For the full topology, exact framing arithmetic gives 40,924 seed-derivation calls, 40,960 leaf commitments, 40,960 tape expansions, and three transcript XOF calls: 122,847 XOF calls in total. Under the selected sponge this expands to 389,974 Anemoi permutations and 131,031,264 permutation nonlinear rows. This count omits non-permutation native rows for extension-field operations, corrections, packing, and wire equalities. It is therefore a lower component count, not a complete π_{issue} benchmark.

The full 18-tree production execution is deliberately opt-in and has not been run for this release. Version 2.5 runs one 2,048-leaf tree shape, but no 18-tree native row stream, complete shard assignment, or parent inter-call wire-identity proof has been produced. Consequently the parent archive still has one named external assertion. Version 2.1 establishes reference semantics and the exact byte ABI; it does not establish native circuit closure, CAP extraction, fork-specific blindness, one-more unforgeability, or the paper’s signature size.

Zero-callback reduced native lowering in version 2.2. The module `pq_rbbc_cap_native.py` replays all 23 reduced CAP XOF calls and one final request-binding call as native Anemoi sponge traces. It links each encoded payload bit to its originating salt, root, parent seed, leaf seed, tape, correction, consistency value, canonical commitment bit, or message bit with ordinary field rows. The canonical 215-byte reduced commitment is an explicit constrained output, and those exact wires feed the 576-bit $H_{\text{RBBC}}(m, c_r)$ trace. Verification evaluates only the resulting rank-one rows; it invokes no CAP or hash callback and records zero external assertions.

The frozen reduced component contains 59,602 wires and 88,282 rows. Of these, 58,462 are nonlinear under the direct model and 29,820 are linear. Its 24 XOF calls require 57 Anemoi permutations, contributing 19,152 permutation rows; the remainder includes payload/output bitness, framing, packing, and 26,848 source-link rows. The 51,845,969-byte canonical JSON row stream has SHA-256

f6a6a0b65e6de16f7bb1d6b42302a12
b004befa62b629d252861e2c986917263.

Changing salt, a GGM root, and the message changes the assignment and outputs but leaves this row-stream digest unchanged. Separate stale-witness tests flip salt, message, commitment, and request-hash bits; every mutation violates an ordinary row.

This closure is intentionally narrow. The reduced witness has only 64 bits, so its polynomial hash occupies one $\mathbb{F}_{2^{193}}$ coefficient and the non-XOF consistency algebra is linear. Production requires a 2,048-bit multi-coefficient polynomial hash and a 2,450-bit tape, exceeding one 772-bit squeeze block. Version 2.2 therefore rejects production parameters. The production parent archive retains its one external assertion.

Production-width multi-squeeze lowering in version 2.3. The direct sponge evaluator already defined ordinary sponge continuation, but the version-2.2 native relation exposed at most one 772-bit rate block. Version 2.3 adds a native permutation between every later squeeze block and links all eight input state lanes to the preceding constrained output state. Each exposed lane is decomposed into 193 binary wires and packed back to the state element; a partial final lane exposes only the requested prefix while still constraining the complete field element. There is no counter or callback: the construction follows the frozen sponge’s state-continuation semantics exactly.

For the deterministic CAP tape input, 2,450 bits require four squeeze blocks and three additional permutations. Together with two absorb permutations, the standalone trace has 4,845 wires and 4,858 rows: 1,680 permutation rows, 568 payload-bitness rows, 2,509 output-bitness rows, and 101 linear rows. Its 3,216,517-byte canonical row stream has SHA-256

c42aec869c6de08695bcb11e2fade158
cc4683aa3ac999e440b78c6ad0314c34.

Flipping a bit in the second output block violates its packing row; flipping an intermediate squeeze state violates the next permutation-input link. Changing the seed while preserving lengths leaves the row stream byte-for-byte unchanged.

The extended CAP fixture keeps two four-leaf trees and a 64-bit witness but sets the tape width to the production value. All eight tape XOF calls therefore exercise the four-block path. The resulting component has 85,034 wires and 113,802 rows, of which 83,510 are nonlinear and 30,292 are linear. Its 24 XOF calls use 81 Anemoi permutations. The 69,273,394-byte row stream has SHA-256

98222b0cafeb944184e3939a878d1e3f
b3af05d10c9795ef0701c87f95462855.

All XOF payloads, the 215-byte canonical commitment, and the final H_{RBBC} join remain ordinary rows with zero external assertions. This is not a security parameter set: the witness still occupies one $\mathbb{F}_{2^{193}}$ coefficient, so it does not test variable-point field multiplication or the production 18-tree topology.

Native field multiplication and Horner hashing in version 2.4. Write the polynomial-basis packing of LSB-first bits as

$$\text{Pack}(b_0, \dots, b_{\ell-1}) = \sum_{j=0}^{\ell-1} b_j X^j \in \mathbb{F}_{2^{193}}, \quad \ell \leq 193.$$

For packed operands A and B , the backend row $A \cdot B = C$ is one native rank-one constraint; reduction modulo the frozen Conway polynomial is the field operation, not an unconstrained host computation. Usability as a bit-facing gadget additionally requires 386 input-bitness rows, 193 output-bitness rows, and one output-pack equality. The standalone fixture therefore has 580 wires and 581 rows. Its product is

011e6dabdb4a6f680c8f194ffd9d0dfc4a7a20a7a5599900c9,

and its 269,342-byte row stream has SHA-256

1c2c2bb9fa869fd43f1bc1f7089e5e0
5a6f9e5cba1d1c8d0a6a80039417681f7.

For a bit vector P , define coefficients c_i by consecutive 193-bit chunks. At point a , the native relation evaluates

$$z \leftarrow c_{q-1}, \quad z \leftarrow za + c_i \quad (i = q - 2, \dots, 0).$$

The 2,048-bit production width gives $q = 11$ coefficients, so two evaluation points require exactly $2(q - 1) = 20$ multiplication rows. The relation also constrains $a_0, a_1 \neq 0$ and $a_0 \neq a_1$ through inverse witnesses:

$$a_0 u_0 = 1, \quad a_1 u_1 = 1, \quad (a_0 + a_1)v = 1.$$

It then bit-decomposes both outputs. The frozen standalone trace has 2,843 wires and 2,845 rows (2,843 nonlinear and two linear), and zero external assertions. Its 1,714,967-byte row stream has SHA-256

0c9d742d44808a20a35838be84a63892
4dc5b2f9183bba731eefba1cb9069850.

Regression tests compare both outputs against the direct CAP polynomial evaluator and reject stale witnesses after changing the vector, either point, an intermediate product, or an output bit. A second valid vector/point pair produces the identical row stream, establishing topology independence for fixed widths.

Symbolic-mask CAP integration in version 2.4. The combined fixture uses a 386-bit witness (two coefficients), two consistency points, two four-leaf trees of extension degree three, and $\rho = 1678$, hence $386 + 1678 + 386 = 2450$ tape bits. The plain vector needs one Horner call. Each of the two trees contributes three extension-bit slices, so the mask algebra needs six more calls. Across seven calls this gives 14 multiplication rows, three point-validation rows, 2,702 output-bitness rows, and 14 output-pack rows.

The complete zero-callback component contains 92,816 wires and 125,401 rows: 91,232 nonlinear and 34,169 linear. Its 24 XOF calls use 84 Anemoi permutations. The 77,156,408-byte row stream has SHA-256

ca391f7d64f649b26c98d646dd1c382a
ebcf848ab849316cb1f65d040c184525,

and the canonical 304-byte commitment has SHA-256

52888b6f229d4e252534d594d0ceaf9b
991be649a1230e269ea6273762499e21.

The native commitment and H_{RBBC} output equal the direct reference. Separate mutations of a plain Horner product, a symbolic-mask Horner product, and an output bit violate ordinary rows. This fixture is intentionally non-secure: it demonstrates the missing algebraic composition but does not replace the 18-tree production trace or establish fork-specific extraction.

Bounded-memory production-tree shard in version 2.5. The first production-shape shard instantiates one of the sixteen 2^{11} -leaf trees. Its witness width is 2,048 bits, so every leaf invokes the eleven-coefficient Horner relation at both transcript-derived points. The degree-12 extension representation contributes twelve symbolic mask slices. Linearity permits the implementation to evaluate each leaf once per point and then aggregate the two field outputs into the plain value and the selected extension slices. It therefore avoids constructing 2,450 coordinate-wide mask vectors in memory.

Only the 2,048 witness wire identifiers and 386 $\widehat{M}$ wire identifiers per leaf are retained in a temporary little-endian spool. The spool is 39,878,656 bytes and has SHA-256

87960a5803e2663a40b3c0bda1611840
806e649f3927c238bd63bdce08812f49.

Every other row is serialized directly into a canonical NDJSON SHA-256 sink. The sink counts 6,146 XOF calls including request binding, 19,505 Anemoi permutations, 40,960 Horner multiplication rows, 28,546 field-aggregation rows, three point-validation rows, and 5,018 final output-bitness rows.

The complete streamed topology contains 19,903,324 wires and 26,126,283 rows: 19,509,254 nonlinear and 6,617,029 linear. The virtual canonical row stream is 18,869,935,441 bytes and has SHA-256

2cfc3641a94635af35dfa5494c61e74a
416ef2fb446975cd417891d244943dfc.

The deterministic one-tree commitment is 206 bytes with SHA-256

14fab7548083411124176bb8e094628f
e6d20347cd78929573b76ab2cd3e757a.

On the recorded run, eight worker processes completed the reference and row stream in 781.76 seconds with peak RSS 140,932 KiB. These are engineering measurements, not prover benchmarks.

Proposition 8.2 (One-shard streamed topology closure). *For the frozen one-tree profile, every GGM, leaf commitment, tape expansion, Horner multiplication, extension-slice aggregation, transcript hash, canonical commitment bit, and request-binding bit is represented by an ordinary native row or a wire equality, and the streamed relation records zero external assertions.*

Proof. The native permutation template is locally satisfiable and is remapped by disjoint wire offsets for every sponge call. Each payload wire is bit constrained and linked to a repeatable source linear form. Each leaf witness is split into eleven consecutive field coefficients; Horner evaluation uses ten native multiplications per point. Field-level xor aggregation is sound by linearity over characteristic two, and the final plain and twelve mask values are decomposed and packed back into constrained bits. H1, the two validated points, H2, canonical serialization, and the request hash consume exactly those wire forms. Canonical serialization counts and hashes every emitted row, and the manifest records an external-assertion count of zero. $\square$

The proposition is intentionally about topology, not full shard witness evaluation or production security. The implementation does not archive a complete 19,903,324-wire assignment, so it cannot yet supply a backend-checked honest assignment or stale-witness mutation proof for the whole shard. The profile also contains only one tree, omits all cross-tree corrections, and is explicitly marked non-secure. Those omissions prevent any inference that the full CAP relation, Blind-UOV fork, or π_{issue} is closed.

The earlier 240/336/360 discrepancy remains documented, but it no longer blocks implementation: version 2.0 intentionally chooses the 336-row profile. The cost of that choice is loss of bit-exact Blind-UOV compatibility and loss of automatic inheritance of its security and signature-size claims.

For historical comparison, version 1.9 used an illustrative paper row as its anchor. The Shockwave/Anemoui Blind-UOV-Is/Shorter result is reported as approximately 16 million R1CS constraints, a 2.8 MB proof, 32 s proving, and 2.5 s verification. The paper does not report the corresponding Blind-UOV-III row. Scaling the native constraint anchor by the witness-width ratio $(576 + 1472)/(256 + 640) = 2.2857$ gives a rough 36.57-million-row native baseline; after the exact 684,419-row non-overlapping ticket/trace increment below, the total is about 37.26 million. The extra 1,152 bitness rows in the portable v2.0 archive are already part of a genuine native π_1 relation and are not double-counted in this combined estimate. This is an extrapolation, not a paper result or local benchmark, and it is not a version-2.0 fork estimate. The fork changes the permutation constraint count and has not yet materialized the complete CAP circuit. The paper explicitly treats these as estimates and notes that the benchmarked implementations do not provide the required zero-knowledge mode. It also does not establish the simulation-extractability required by our concurrent GCCA proof. Therefore Shockwave/Anemoui is a *costing backend*, not yet the security-frozen Π_{issue} implementation [1].

The multivariate commitment $c = (\mu + F(r), G(r))$ and relation proofs of Bouillaguet et al. are a promising later optimization because they are algebraically aligned with UOV, but they add a newer MQ/bilinear-system assumption and are not the baseline for the first auditable construction [8].

8.2 Threshold Niederreiter for trace encryption

Code and plaintext profile. Let $\mathcal{G}$ be a binary Goppa code of length n , dimension k , and designed error weight t , with public systematic parity-check matrix $H_{\text{pub}} = (I_{n-k} \mid T)$. The selected decoder MUST guarantee $d_{\min}(\mathcal{G}) \geq 2t + 1$. Fix $\text{rid} \in \{0, 1\}^{256}$ and $\text{sn} \in \{0, 1\}^{128}$, and define the fixed 48-byte plaintext

$$\mu = \text{Encode}_{48}(\text{rid} \parallel \text{sn}).$$

The online ciphertext uses a weight- t error $e \in \mathbb{F}_2^n$ as its only encryption witness. Thus the abstract randomness ρ_{tr} is instantiated by e .

Hash DEM. Sample e uniformly from the weight- t vectors and compute

$$s = H_{\text{pub}} e^{\top} \in \mathbb{F}_2^{n-k}, \quad \text{wt}(e) = t, \quad (\text{N1})$$

$$\text{ad} = \text{Encode}(\text{ctx}, \text{sn}, h), \quad (\text{N2})$$

$$Z = \text{SHAKE256}(\text{PQ-RBBC}/\text{KDF} \parallel e \parallel s \parallel \text{ctx}, 640), \quad (\text{N3})$$

$$(K_{\text{mac}}, P) = Z, \quad |K_{\text{mac}}| = 32, \quad |P| = 48, \quad (\text{N4})$$

$$c_{\mu} = \mu \oplus P, \quad (\text{N5})$$

$$\gamma = \text{KMAC256}_{K_{\text{mac}}}(s \parallel c_{\mu} \parallel \text{ad}, 256; \text{custom} = \text{PQ-RBBC}/\text{TAG}), \quad (\text{N6})$$

$$C = (s, c_{\mu}, \gamma). \quad (\text{N7})$$

The output length in (N3) is in bits. SHAKE256 and KMAC256 are the standardized extendable-output and MAC functions of FIPS 202 and SP 800-185, respectively [13, 14]. The composition HDEM is nevertheless a new protocol profile, not a standardized AEAD and not yet a production claim. A standardized one-time AEAD may replace it later without changing the abstract R_{enc} interface.

Definition 8.3 (Concrete trace relation). For $C = (s, c_{\mu}, \gamma)$, define

$$R_{\text{Ntr}}(\text{tpk}, C, \text{ctx}, h; \text{rid}, \text{sn}, e) = 1$$

iff tpk selects H_{pub} and the fixed profile above, and Equations (N1)–(N7) all hold with $\mu = \text{Encode}_{48}(\text{rid}||\text{sn})$. In this version,

$$\begin{aligned} R_{\text{enc}}(\text{tpk}, C, \text{Encode}(\text{ctx}, \text{sn}, h); \text{rid}||\text{sn}, \rho_{\text{tr}}) &= 1 \\ \iff R_{\text{Ntr}}(\text{tpk}, C, \text{ctx}, h; \text{rid}, \text{sn}, e) &= 1, \quad \rho_{\text{tr}} = e. \end{aligned}$$

Hence $\mathcal{C}_{\text{trace}}$ is now the exact conjunction (N1)–(N7), not an unspecified “encrypt correctly” box.

Lemma 8.4 (Unique syndrome witness). *For a code with $d_{\min} \geq 2t + 1$, a syndrome s has at most one witness e of weight at most t satisfying $s = H_{\text{pub}}e^\top$.*

Proof. If e and e' were two such witnesses, then $d = e + e'$ would satisfy $H_{\text{pub}}d^\top = 0$ and $\text{wt}(d) \leq 2t$. If $e \neq e'$, d is a nonzero codeword below the minimum distance, a contradiction. Therefore $e = e'$. This is the standard unique-decoding argument used by Classic McEliece [11]. $\square$

Corollary 8.5 (Unique trace opening). *The concrete relation R_{Ntr} has information-theoretically unique plaintext openings.*

Proof. For a fixed C , Lemma 8.4 fixes e from s . Equations (N3)–(N4) then deterministically fix P and K_{mac} , and (N5) fixes the sole $\mu = c_\mu \oplus P$. The fixed-width encoding is injective, so it fixes (rid, sn) . Equation (N6) additionally rejects any inconsistent associated data. No computational assumption is needed for uniqueness. $\square$

Proposition 8.6 (Concrete trace correctness). *If threshold decoding reconstructs the unique error e for s , the combiner recomputes (N3)–(N6), returns $\mu = c_\mu \oplus P$, and accepts only if γ verifies and the decoded serial equals the visible sn . Consequently every honestly formed concrete trace ciphertext opens to exactly $\text{rid}||\text{sn}$.*

Privacy assumption and proof boundary. Let t-OW-SD be the game in which an adversary receives H_{pub} , fewer than τ threshold shares, and $s = H_{\text{pub}}e^\top$ for uniform weight- t e , and must output e . We assume this advantage is negligible for the frozen distributed decoder.

Theorem 8.7 (Conditional concrete trace privacy). *Model SHAKE256 as a quantum random oracle, assume KMAC256 is a quantum-secure PRF, never reuse e , and let $\epsilon_{\text{O2H}}(q_H)$ denote the loss of the applicable one-way-to-hiding lemma for q_H quantum hash queries. Then for every fewer-than-threshold QPT distinguisher $\mathcal{A}$ there are QPT reductions $\mathcal{B}_1, \mathcal{B}_2$ such that*

$$\text{Adv}_{\text{Ntr}, \mathcal{A}}^{\text{t-IND-CPA}} \leq \text{Adv}_{\mathcal{G}, \mathcal{B}_1}^{\text{t-OW-SD}} + \epsilon_{\text{O2H}}(q_H) + \text{Adv}_{\text{KMAC256}, \mathcal{B}_2}^{\text{qPRF}}.$$

Proof. Starting from the real game, replace the 640-bit SHAKE256 value at the hidden input (e, s, ctx) by a uniform string. A distinguisher for this step either yields a threshold syndrome-decoding solver or is bounded by the stated one-way-to-hiding loss. Now P is an independent uniform 48-byte pad and K_{mac} is an independent uniform key. Replace KMAC256 under that key by a random function, paying the qPRF term. The value $c_\mu = \mu_b \oplus P$ and the tag are then independent of b , so the final advantage is zero. Summing the hybrid differences gives the bound. $\square$

A new accepted tuple (s, c'_μ, γ') with the same hidden derived key and different authenticated input gives a KMAC forgery, up to the same syndrome-decoding/QROM bad event. Thus the xor pad gives one-time privacy and KMAC256 gives one-time ciphertext integrity. Substituting Theorem 8.7 for the abstract τ -IND-CPA term in Theorem 7.11 yields protocol-level τ -GCCA, plus the SE-AoK, forked blind-signature, CAP, unique-opening, authorization, and hash terms already displayed there. This does *not* prove standard IND-CCA security for a bare ciphertext endpoint.

The threshold-Niederreiter work solves malformed-ciphertext attacks by attaching a straight-line-extractable validity proof to every public ciphertext; its authors report about 30.5 KB for a three-party ciphertext, with the proof dominating the syndrome [9, 10]. Here the same validity language $\{s : \exists e, \text{wt}(e) = t, s = He^\top\}$ is certified once inside π_{issue} . The online ticket retains only $C = (s, c_\mu, \gamma)$, and the raw distributed decoder remains unreachable.

Exact online byte counts. Because $|c_\mu| = 48$ and $|\gamma| = 32$, the trace ciphertext is exactly the syndrome length plus 80 bytes. Using the published Classic McEliece implementation sizes gives the following anchors [12]:

Parameter anchor	Public key	Syndrome	This C_{trace}
mceliece348864	261120 B	96 B	176 B
mceliece460896	524160 B	156 B	236 B
mceliece6688128	1044992 B	208 B	288 B
mceliece6960119	1047319 B	194 B	274 B
mceliece8192128	1357824 B	208 B	288 B

The aggressive working anchor is mceliece6688128: approximately 1.045 MB of cached public key buys a 288-byte trace ciphertext. The 368-byte payload is exact. A 12,012-byte complete ticket results only if the fork reproduces the paper’s 11,644-byte signature target; version 2.4 has not established that condition.

Remark 8.8 (Mandatory 2026 cryptanalytic hold). *The anchor above is not frozen. A 10 August 2026 preprint gives a classical quasipolynomial-time distinguisher applying to all Classic McEliece NIST parameter sets and extends it to a heuristic quasipolynomial-time decryption attack; its concrete improvements are described as not yet practical [15]. That is not a demonstrated practical break, but it materially changes confidence. Deployment parameter selection is blocked until independent review quantifies this attack for the threshold construction.*

8.3 Reproducible five-block circuit audit

Counting model. The paper’s Anemoi profile uses an R1CS over $\mathbb{F}_{2^{129}}$ for NIST level I. Addition, xor, multiplication by public constants, fixed-width concatenation, and the public binary syndrome map are linear. Each Boolean AND or field multiplication contributes one nonlinear R1CS constraint. Following the Blind-UOV paper, one Keccak-f permutation costs $24 \cdot 1600 = 38,400$ constraints [1]. The audit counts only constraints added beyond a native well-built-request π_1 relation. The complete PQ-RBBC-BUOV-336 CAP circuit is not yet materialized, so the historical paper-anchored level-III baseline below is an explicit version-1.9 comparator, not a version-2.5 fork estimate.

Exact-weight gadget. The witness e has 6,688 bits. Pad it with 1,504 public zeros to 8,192 inputs and sum them using a fixed balanced binary adder tree. Combining two j -bit counters uses one half-adder and $j - 1$ full adders. Since xor is linear, the nonlinear cost is $1 + 2(j - 1) = 2j - 1$. The exact cost is therefore

$$\sum_{j=1}^{13} \frac{8192}{2^j} (2j - 1) = 24,547$$

AND constraints. Equating the final 14-bit count to 128 is linear. This gadget avoids the invalid shortcut $\sum e_i = 128$ in characteristic two, which would check only parity. Separately, e , sn , and k_{hold} require $6688 + 128 + 256 = 7,072$ bitness constraints.

Exact Keccak accounting. SHAKE256 has a 136-byte rate. With pad_{10^*1} , an L -byte message uses $\lfloor L/136 \rfloor + 1$ absorb permutations. The fixed encodings give:

Invocation	Absorbed bytes	Keccak-f calls	Nonlinear constraints
$H_{\text{ticket}}(M)$	$14 + 368 = 382$	3	115,200
$H_{\text{hold}}(k)$	$12 + 32 = 44$	1	38,400
Trace SHAKE KDF	$11 + 836 + 208 + 32 = 1087$	8	307,200
KMAC256	$136 + 136 + 336 + 3 = 611$	5	192,000
Total		17	652,800

For KMAC, the first 136-byte block is the cSHAKE function/customization prefix, the second is $\text{bytepad}(\text{encode_string}(K), 136)$, the authenticated data is $208 + 48 + 80 = 336$ bytes, and $\text{right_encode}(256)$ is three bytes. All squeezing fits in the first rate block and adds no permutation.

Five-block result. The incremental nonlinear counts are:

Block	Constraints	Contents
C_{shape}	128	serial bitness; widths/wiring linear
$C_{\text{ticketHash}}$	115,200	three SHAKE permutations
C_{RBBCmask}	1,152	bitness of r, h_{RBBC} ; 576 mask equations linear
C_{holder}	38,656	256 bitness + one SHAKE permutation
C_{trace}	530,435	e bitness, weight, KDF, and KMAC
Portable v2.5 relation	685,571	includes 1,152 native-overlap hardening rows

Of the portable total, 1,152 bitness rows protect the explicit r and h_{RBBC} interface while the full native generator is absent. A genuine native π_1 already constrains those binary values, so only 684,419 rows are non-overlapping additions to the paper-anchored native baseline. The 576 $y = r + h_{\text{RBBC}}$ equations are linear. The syndrome equation adds 1,664 linear output equalities but no nonlinear gate. The xor pad, associated-data construction, external/internal serial equality, and final constant comparisons are also linear.

Historical paper-anchored issuance estimate. The level-III native baseline is estimated from the paper’s rounded 16-million-constraint NIST-I figure by the witness-width ratio:

$$N_{\text{native,III}} \approx 16,000,000 \frac{576 + 1472}{256 + 640} = 36,571,429.$$

Adding the exact incremental relation gives

$$N_{\pi_{\text{issue}}} \approx 36,571,429 + 684,419 = 37,255,848,$$

or 2.3285 times the NIST-I anchor. Applying the paper’s square-root proof-size scaling and linear time scaling gives a rough 4.27 MB proof, 74.5 s proving, and 5.82 s verification. None of these level-III figures is a paper result or local benchmark; the scaling may omit parameter-dependent costs. Peak prover memory remains a measurement obligation.

Version-2.1 CAP lower component count. The direct fork accounting supersedes the historical 36.57-million-row CAP anchor as an engineering estimate. Its sponge calls alone require 131,031,264 nonlinear permutation rows before extension-field arithmetic, packing, correction, and equality rows are added. Adding only the exact 684,419 non-overlapping ticket/trace increment already reaches 131,715,683 rows, still excluding those omitted CAP rows. Therefore the earlier 37.26 million, 4.27 MB, and 74.5 s extrapolations are retained only as historical paper comparisons and must not be used to budget this fork. The direct profile is aggressive in online bytes but computationally heavy for offline π_{issue} .

Proposition 8.9 (Provisional monolithic-circuit feasibility decision). *Under the historical version-1.9 costing model above, adding the entire ticket and trace relation to the estimated well-built-request NIZK changes its baseline by about 1.87%. This percentage must be recomputed after the full forked CAP circuit exists. Until then, there is no demonstrated communication-based reason to split R_{aug} into linked proofs at this stage. A split is justified only if the selected simulation-extractable backend, implementation memory, or prover parallelism makes the monolithic circuit impractical.*

Exact payload and provisional complete-ticket target. For mceliece6688128, $|C| = 288$ bytes and

$$|M| = 32 + 16 + 32 + 288 = 368 \text{ bytes.}$$

Combining this with the published 11,644-byte Blind-UOV-III/Shorter signature as a provisional fork target gives

$$|T| = |M| + |\sigma| = 12,012 \text{ bytes}$$

before transport framing. This is conditional arithmetic, not a measured version-2.5 signature size. Under the same provisional assumption, cached public material is approximately 1,044,992 bytes for the trace key plus the paper’s rounded 189.2 KB Blind-UOV verification key. Relative to a 70 KB online credential, the target is about 5.83 times smaller. The roughly 4.27 MB issuance-NIZK figure is a superseded historical comparator; version 2.5 already counts more than 131 million CAP permutation rows. Both the CAP commitment and π_{issue} remain offline and are never forwarded to a satellite verifier.

Remark 8.10 (Audit reproducibility). *The companion executable relation recomputes every exact incremental integer above and records it in the version-2.5 manifest. Fork signature and native CAP estimates are kept in separately labelled fields so a rounded extrapolation cannot be mistaken for a benchmark.*

8.4 Executable reference relation and rejection tests

Version 2.5 preserves and audits the incremental relation in `pq_rbbc_reference.py`. The witness builder derives h , s , c_μ , γ , the canonical 368-byte payload, and its ticket digest from $(\text{rid}, \text{ctx}, \text{sn}, k_{\text{hold}}, e)$. The circuit builder then emits a stream of characteristic-two events of four kinds: linear wire definitions, linear equality assertions, Boolean multiplication gates, and bitness constraints $x(x + 1) = 0$. A backend can consume the same stream without retaining the whole circuit in memory. This version executes the stream and counts it, but does not yet flatten it into a proof system’s matrix format.

The symbolic Keccak-f implementation performs all 24 rounds. Its SHAKE256 output is checked against Python’s independent FIPS 202 implementation, and the SP 800-185 KMAC256 output is checked against the OpenSSL 3 KMAC provider [13, 14]. For the frozen honest vector the generated circuit reports:

Measured generator quantity	Value
Allocated wire identifiers	2,980,304
Linear definitions	2,282,475
Linear equality assertions	3,534
Nonlinear constraints	685,571
Public input bits	4,032
Secret input bits	8,224
External forked CAP-hash assertions	1
Keccak-f permutations	17
Failed assertions	0

Thus the executable result exactly reproduces every nonlinear block total in the preceding audit. The linear-definition figure is diagnostic IR size, not an additional nonlinear count: the costing model requires a backend to inline affine expressions. A backend that instead materializes each affine wire as an R1CS row must report that larger total explicitly. On the present reference environment, one honest streaming generation takes about six seconds. This is Python relation generation time, not NIZK proving time, and no prover-memory claim follows from it.

The regression suite accepts the honest witness and sends eleven independent mutations through the complete circuit: wrong error weight, changed syndrome, changed masked identity, changed holder hash, changed KMAC tag, changed serial, changed blind request, changed blind mask, changed H_{RBBC} image, changed CAP randomness, and changed common context. All eleven are rejected. The checked output is recorded in `pq_rbbc_reference_manifest_v2_2.json`.

Exact implementation boundary. The $H = (I \mid T)$ matrix in the reference vector is generated deterministically from a public seed and is only a rank-controlled test fixture; it is not a certified binary Goppa key. Similarly, `TestPQRBBC336Adapter` uses the selected Anemoi sponge for H_{RBBC} but derives a 48-byte commitment with a labelled SHAKE test fixture. It is not a blind signature or production CAP. The new CAP module and strict API separately test the production serialization and byte join, but this incremental relation still marks one native full-row CAP-plus-hash boundary. The 576 public y bits and 1,152 secret r, h_{RBBC} bits are serialized as circuit inputs; m is derived internally, and ρ, c_r remain hidden behind the native boundary. Consequently this implementation establishes that the incremental relation is internally consistent and executable; it does not establish that the combined native fork proof is zero knowledge or simulation extractable.

8.5 Portable binary R1CS lowering

Version 2.3 preserves the proof-backend exchange layer in `pq_rbbc_br1cs.py`. It consumes the same event stream as the counting sink and serializes three standard rank-one forms over $\mathbb{F}_2$:

$$\begin{aligned} \text{MUL}(a, b, c) : \quad & a \cdot b = c, \\ \text{LINDEF}(c; x_1, \dots, x_j; k) : \quad & (c + x_1 + \dots + x_j + k) \cdot 1 = 0, \\ \text{ASSERT}(x_1, \dots, x_j; k) : \quad & (x_1 + \dots + x_j + k) \cdot 1 = 0. \end{aligned}$$

Wire identifiers and arities use unsigned LEB128 encodings. A fixed 124-byte header commits to all row counts, the body length, and the SHA-256 digest of the body. Public and secret input declarations contain identifiers and visibility only; assignment values are not embedded in the circuit archive.

The resulting honest archive has the following exact dimensions:

Portable backend quantity	Value
Wires	2,980,304
Public input bits	4,032
Secret input bits	8,224
Nonlinear R1CS rows	685,571
Materialized affine-definition rows	2,282,475
Linear assertion rows	3,534
Total R1CS rows	2,971,580
Archive size	49,227,687 bytes

The 685,571 figure is therefore a *nonlinear-only* cost under the affine elimination model used in Section 8.3. The 2,971,580 figure is the portable total when every affine definition is materialized as its own rank-one row. They are not competing measurements. A concrete proof backend may

recover the smaller representation by substituting affine expressions into multiplication rows, but that optimization requires its own soundness argument and resource benchmark.

Although the exchange format names $\mathbb{F}_2$ as its base relation, the same Boolean assignment and equations embed into $\mathbb{F}_{2^d}$: addition remains xor, and the input bitness equations prevent non-Boolean witnesses. This makes the archive compatible in principle with both the $\mathbb{F}_{2^{129}}$ costing anchor and the target $\mathbb{F}_{2^{193}}$ level-III field, without claiming that the required lift or a specific prover has been integrated.

An independent reader reconstructs and evaluates all 2,971,580 rows. It accepts the honest assignment, rejects a one-bit assignment mutation, and rejects a one-bit archive corruption through the body digest. Lowering a wrong-weight witness produces exactly the same 49,227,563-byte body and the same digest as the honest witness, showing that circuit structure is witness-independent; its assignment nevertheless fails 320 rows. Five end-to-end back-end tests pass; wall time is environment-dependent. The checked metadata is recorded in `pq_rbbc_br1cs_manifest_v2_5.json`.

Remaining external rows. The archive records one named external assertion for the full 18-tree native row stream and exact H_{RBBC} wire join. Direct reference evaluation and the canonical byte join exist, but the archive cannot yet evaluate those operations as one native witness-independent row stream. The linear equation $y = r + h_{\text{RBBC}}$ is no longer external. This narrowed boundary is deliberate and machine-visible: an archive with an unchecked external assertion is not a closed issuance proof. The next integration must replace that marker with the forked CAP.Commit constraints and exact sponge wire identities; deleting or treating it as automatically true is forbidden.

8.6 Local mixing as a future trace-encryption candidate

The core does not depend on elliptic curves, lattices, or codes. A future local-mixing encryption may replace threshold Niederreiter only if it supplies all four properties (T1)–(T4), an efficiently provable NP encryption relation, and a practical τ -party opening protocol. A small standalone ciphertext claim is insufficient by itself: threshold opening, unique opening, and extractable issuance binding are the missing system properties.

9 What has and has not been proved

Status	Result or obligation
Conditional reductions	Correctness; proof/request/message binding; certified ciphertext validity; signature-gated decoder safety; issuer unlinkability against an honest-protocol curious issuer; trace soundness; one-more ticket unforgeability; and τ -GCCA trace privacy.
Assumed component property	Post-quantum SE-NIZK AoK; CAP straightline extraction and unique committed mask; QROM cross-message request binding; PQ-RBBC-BUOV-336 fork blindness and one-more unforgeability; threshold syndrome-decoding privacy with robust shares; QROM one-way-to-hiding; KMAC integrity; and authorization unforgeability.

Status	Result or obligation
Concretely instantiated	A 72-byte hidden-state request ABI; executable five-block augmented relation; exact Niederreiter–SHAKE–KMAC trace relation; information-theoretic unique opening; independently checked SHAKE/KMAC; full-circuit positive and negative tests; an internally constrained $y = r + h_{\text{RBBC}}$ equation; a fail-closed fork-native import contract; a check-summed portable R1CS archive with independent row evaluation; 685,571 incremental nonlinear rows; 2,971,580 materialized total rows; a source-pinned $\mathbb{F}_{2^{193}}$ 336-row Anemoi permutation; and an independent 772-bit rate/772-bit capacity sponge with arbitrary-length native squeezing and a 576-bit request output. The frozen two-block hash trace has 2,238 rows and 2,235 wires; a direct CAP.Commit reference with salted GGM trees, leaf commitments/tapes, corrections, and consistency hashing; a canonical 5,378-byte production commitment encoding; a strict byte-level CAP-to-H_{RBBC} join; an 88,282-row zero-callback reduced native trace; a 4,858-row standalone 2,450-bit tape trace; a 113,802-row extended CAP trace; a 2,845-row production-width 11-coefficient Horner trace; a 125,401-row two-coefficient CAP/Horner integration whose eight tapes have production width and a bounded-memory one-tree production-shape stream with 26,126,283 rows, 19,903,324 wires, the full 2,048-bit witness, degree-12 masks, two points, and 2,450-bit tapes (all CAP fixtures non-secure and zero external assertions; the one-tree run freezes topology but not a complete assignment archive).
Not yet instantiated	A post-quantum zero-knowledge and simulation-extractable implementation of Π_{issue}; a native $\mathbb{F}_{2^{193}}$ materialization of the full 18-tree CAP row stream and exact inter-call wire identities; a complete assignment or backend-linked witness for each streamed shard; the 2^{12}-leaf shard; execution of a full 18-tree production vector; a fork-specific blindness/one-more-unforgeability proof; fresh key, signature, and proof-size benchmarks; validation of the CAP properties used by Theorem 3.2; proof-system-specific affine elimination; import of a real Goppa key; peak prover memory; robust threshold-share transcript; proof benchmarks; and a reviewed response to the August 2026 McEliece cryptanalysis.
Explicitly not claimed	Standard IND-CCA security of the bare ciphertext, security against a malicious issuer, metadata/timing anonymity without system controls, UC composability, side-channel resistance, satellite AKE security, or FLEO/handover security.

10 Implementation invariants

The following invariants are part of the cryptographic design and should be machine-tested or enforced by type/API boundaries.

- INV-1 There is no production function that accepts a bare trace ciphertext and returns a partial decryption.
- INV-2 The bytes hashed for the issuer signature are exactly the canonical bytes parsed by ticket verification and opening.
- INV-3 The issuer signs only after successful proof verification and records a completed session only after returning its blind response.
- INV-4 The blind request proven in Π_{issue} is byte-for-byte the request passed to the blind-signing implementation.
- INV-5 The public issuance request contains exactly the 72-byte value y ; it contains neither m nor c_r . The mask and commitment randomness remain hidden and the commit-before-hash order is enforced.
- INV-6 The opening authorization includes the ticket digest and evidence digest, and replay is statefully rejected.
- INV-7 The reconstructed serial number must equal the clear serial number.
- INV-8 Issuer keys and opening keys have independent generation ceremonies, storage, rotation, and compromise domains.
- INV-9 Epoch metadata is federation-fixed; the issuer cannot personalize it.
- INV-10 Every component and parameter set has an explicit post-quantum security claim at the chosen target category.
- INV-11 The parser accepts exactly the frozen 368-byte payload layout and rejects variable-length or alternate encodings before signature verification.
- INV-12 The issuance circuit enforces bitness of all 6,688 error coordinates and proves exact weight 128 with the specified counter gadget; a parity-only check is forbidden.
- INV-13 The Shockwave/Anemoi figures may be used for costing only until a backend meeting the stated zero-knowledge and simulation-extractability model is proved or substituted.
- INV-14 Production configuration rejects the deterministic parity-check matrix and test-only blind-signature adapter; neither fixture may cross the backend boundary.
- INV-15 Backend lowering must preserve every multiplication and equality event, eliminate affine wires soundly, and accept/reject the published reference and mutation vectors identically to the executable IR.
- INV-16 A serialized circuit is accepted only if its declared length, row counts, sequential wire discipline, and body digest all verify before proof generation or verification.
- INV-17 Any nonzero external-assertion count makes the archive incomplete for production issuance; the PQ-RBBC-BUOV-336 marker must be replaced, never ignored.
- INV-18 The 576 mask bits and 576 H_{RBBC} -image bits are secret circuit inputs with bitness constraints; all 576 equations $y_j + r_j + h_{\text{RBBC},j} = 0$ must remain ordinary rows after lowering.

- INV-19 A native import is accepted only if it matches the frozen $\mathbb{F}_{2^{193}}$ profile, has witness-independent topology and zero external assertions, and satisfies every check in the native-import contract.
- INV-20 The selected sponge, version-2.1 CAP reference, version-2.2 reduced native trace, version-2.3 extended multi-squeeze trace, version-2.4 Horner relations, and version-2.5 2,048-leaf streamed shard define the implemented H_{RBBC} wires unambiguously. The CAP fixtures are explicitly non-secure and cannot satisfy the production import contract; a full shard assignment, the 4,096-leaf shard, all 18 trees, and exact parent-circuit wire identities are still required.
- INV-21 The fork must never claim bit-exact Blind-UOV compatibility. The reported 240 rows remain comparison data; the selected fork uses 336 rows.
- INV-22 Production closure requires independent fork vectors and cryptanalysis, a fork-specific blindness/one-more-unforgeability proof, and fresh key, signature, proof-size, time, and memory benchmarks. The paper’s 11,644-byte signature and the derived 12,012-byte ticket remain provisional targets.

11 Next proof step

The independent 336-row permutation, canonical sponge framing, 576-bit H_{RBBC} output, native hash rows, hidden-state adapter, direct CAP reference, canonical 5,378-byte commitment, strict byte ABI, incremental relation, and fail-closed fork contract are complete. The reduced CAP fixture and final H_{RBBC} join also have a zero-callback native row stream with zero external assertions. Generic native multi-squeeze, a frozen 2,450-bit tape, an extended CAP integration, bit-bound native multiplication, production-width Horner hashing, and a two-coefficient symbolic-mask CAP integration are complete as well. The full 2,048-bit witness and every degree-12 symbolic mask slice have now been executed in one real 2,048-leaf production tree shape, with a frozen bounded-memory row-stream digest and zero external assertions. The 240-row artifact gap is no longer an engineering blocker because the fork intentionally gives up bit-exact compatibility. The next component is assignment-backed production composition:

Give the streamed 2,048-leaf shard a complete backend-linked witness assignment and whole-shard stale-witness mutation tests. Add the 4,096-leaf/degree-13 shard, then compose all eighteen trees and their cross-tree corrections under one canonical transcript. The final parent archive must reject changed m , r , ρ , c_r , or hash output and reach zero external assertions. Then prove unique-mask and straightline extraction, revalidate blindness and one-more unforgeability, benchmark the actual signature, and qualify a post-quantum zero-knowledge and simulation-extractable backend.

The 131-million-row permutation subtotal is large enough that monolithic versus linked proof architecture must now be benchmarked; the earlier monolithic preference is no longer justified by the historical 37-million-row estimate. The Niederreiter deployment parameter MUST remain provisional under Remark 8.8. After backend qualification, the remaining cryptographic component is the robust threshold-share transcript. Satellite AKE integration remains out of scope until those two obligations close.

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